

Physiological Synchrony and Collaboration: an empirical inquiry using the Overcooked game

Media Technology Thesis Defence

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Introduction

Physiological Synchrony (PS) refers to the temporal alignment of physiological responses between individuals during social interaction, over time (Helm et al., 2018).

Significance of studying PS

- PS has been studied within offline social interactions
- Higher PS improves collaboration in offline settings, like classrooms (Ahonen et al., 2016).
- In VR, higher PS correlates with better team performance (Moharana et al., 2023).
- Within PS research, a limitation identified was the lack of familiarity between participants (Gordon et al., 2020).

Research gap:

- Limited exploration of PS in virtual environments, specifically virtual gaming environments such as Overcooked.
- Limited exploration on the influence of familiarity on PS in short-term, virtual collaborative settings.

Research Question & Hypotheses

Research Question:

This study sought to investigate the relationship between PS and collaboration among dyads in a virtual gaming environment, such as Overcooked.

Additionally, the study examines the role of familiarity in influencing PS and collaboration.

Hypotheses:

- Higher PS correlates to greater collaboration within dyads.
- Familiar dyads will show higher PS and collaboration.

Methodology (1)

Research Design:

- Quasi-experimental design with dyads playing Overcooked.
- Participants paired as familiar or unfamiliar dyads.
- Heart rate data collected using PPG sensors to measure PS.

Participants:

- 14 participants formed into 7 dyads (3 familiar, 4 unfamiliar).
- Convenience sampling method used for recruitment

Setup:

- Overcooked game used to measure collaboration.
- PPG sensors used to collect heart rate data.
- Collaboration measured via game score.
- PS calculated using cross-correlation of heart rate data from dyads.

Methodology (2)

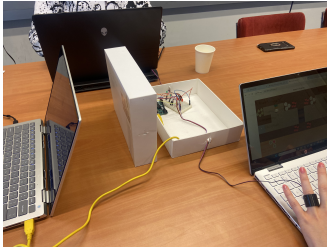


Figure: Dyads engaged in the experiment

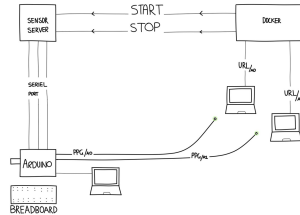


Figure: illustration of the experiment setup

Descriptive statistics

- 14 participants formed 7 dyads.
- Age range: 23-28 years; average age: 26 years.
- Average age difference within dyads: 2 years; max difference: 4 years in 3 dyads.
- Gender distribution: 7 males, 7 females; 2 same-gender dyads, 5 mixed-gender dyads.
- Familiarity: 4 unfamiliar dyads, 3 familiar dyads.
- No missing data; 42 data points collected.

Key Findings: PS & Collaboration

A weak but positive correlation ($r = 0.145$) between PS (Maximum IBI correlations) and collaboration (game score) was observed, though **not statistically significant** with a p-value of ($p = 0.361$).

Variables	n	Pearson's r	p
game_order vs game_score	42	0.133	0.401
game_order vs max_corr_ibi	42	-0.077	0.628
game_score vs max_corr_ibi	42	0.145	0.361

Table: Pearson's correlation coefficients between Game Order, Game Score, and Maximum Correlations of IBI

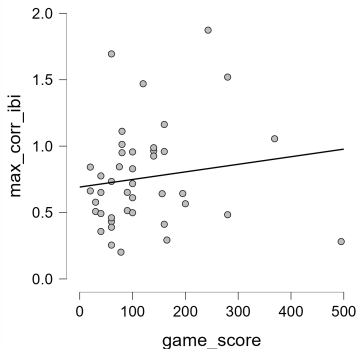


Figure: Distribution of Game Score vs Maximum Correlation of IBI

Key Findings: Familiarity Analysis

Influence of familiarity on PS and Collaboration (Independent T-Test)

- t-values are relatively small ($t = 0.555, 0.493$)
- p-values ($p = 0.582, 0.624$) are greater than 0.05, hence maximum IBI correlation and game scores between familiar & unfamiliar dyads are not statistically significant.

	t	df	p	C.d	SE C.d
max_corr_ibi	0.555	40	0.582	0.173	0.313
game_score	0.493	40	0.624	0.154	0.313

*C.d = Cohen's D

Table: T-test for Familiarity on Game Score and Maximum IBI Correlations

Other findings: - Dyad gender an PS: Same-gender dyads showed higher PS than mixed-gender pairs. - Dyad gender on collaboration (game scores): not statistically significant.

Limitations & Future Work

Limitations:

- Small sample size (14 participants)
- Short game duration
- Only heart rate used to measure PS

Future Work:

- Future studies should include larger samples and utilize more than one physiological measure simultaneously.
- The experimental setup allows easy integration of different games, allowing researchers to further explore PS within different game environments.
- Additionally, the experimental setup can be used to measure PS with PPG sensors and ECGs.

Conclusion

Main Takeaways:

- Research: relationship between PS and collaboration within dyads in a virtual gaming environment using the game overcooked.
- There is a weak, yet positive correlation between PS and collaboration, although not statistically significant.
- Familiarity did not significantly affect PS or game scores.
- Mixed-gender dyads presented lower PS compared to same-gender dyads, but dyad gender did not significantly affect collaboration.

Questions?

References

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